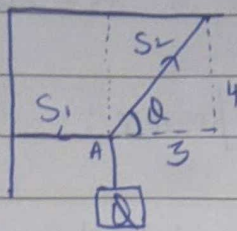


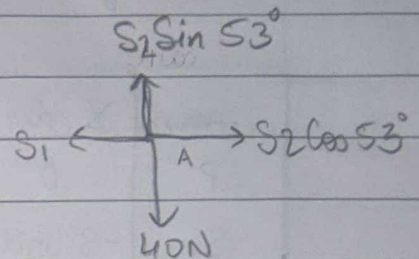
ME Assignment - 3

①



$$\tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

FBD of point A



Force	x-dir	y-dir
S_1	$-S_1$	0
S_2	$S_2 \cos 53^\circ$	$S_2 \sin 53^\circ$
Q	0	-40

Balancing Forces Vertically:

$$40 = S_2 \sin 53$$

$$S_2 = 50 \text{ N}$$

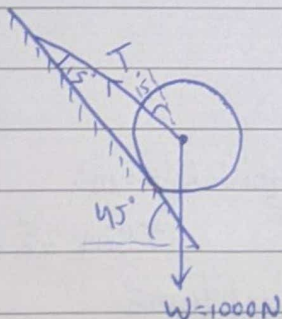
Balancing Forces Horizontally:

$$S_1 = S_2 \cos 53$$

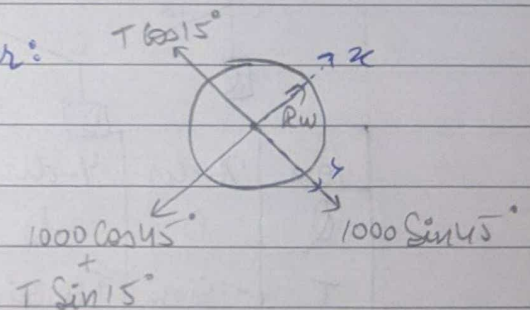
$$S_1 = 50 \cos 53$$

$$S_1 = 30 \text{ N}$$

②



FBD of roller:



Force	x-dir	y-dir
T	$-T \sin 15$	$-T \cos 15$
W	$-W \cos 45$	$W \sin 45$

Balancing forces in ~~y~~ y-dir:

$$1000 \sin 45^\circ = T \cos 15^\circ$$

$$T \approx 732 \text{ N}$$

Balancing forces in x-dir:

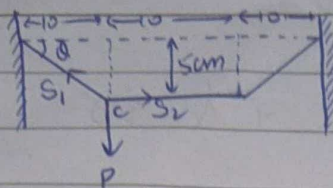
$$R_w = 1000 \cos 45^\circ + 732 \sin 15^\circ$$

NAME - ROHIT KUMAR

SCHOLAR ID - 2015053

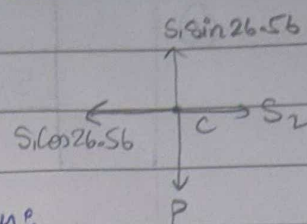
$$R_w = 897 \text{ N}$$

③



$$\tan \theta = \frac{5}{10} \Rightarrow \theta = 26.56^\circ$$

FBD at point C:



Force	X-dir	Y-dir
P	0	-P
S ₁	-S ₁ cos 26.56	S ₁ sin 26.56
S ₂	S ₂	0

Balancing forces vertically:

$$S_1 \sin 26.56 = P$$

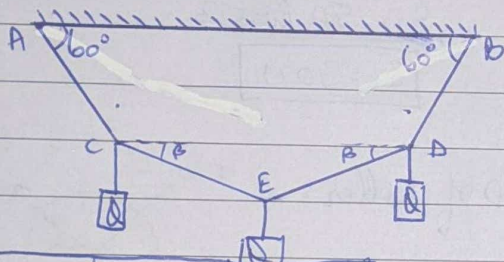
$$S_1 = 2.023 P$$

Balancing forces horizontally:

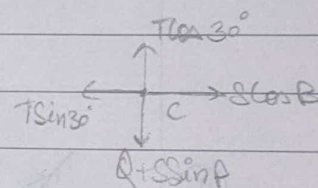
$$S_2 = S_1 \cos 26.56$$

$$S_2 \approx 2P$$

④



FBD of point C:



Force	X-dir	Y-dir
Q	0	-Q
T	-T sin 30	T cos 30
S	S cos beta	-S sin beta

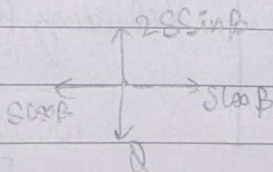
Balancing Horizontal forces:

$$T \sin 30 = S \cos \beta \quad (1)$$

Balancing Vertical forces:

$$Q + S \sin \beta = T \cos 30 \quad (2)$$

FBD of point E:



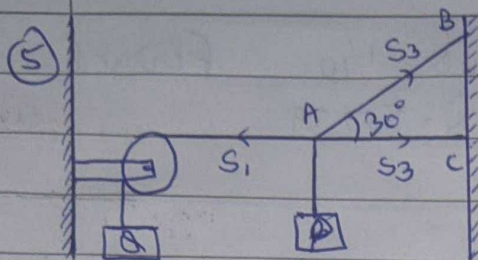
Force	X-dir	Y-dir
Q	0	-Q
S	+S cos beta	2S sin beta

Balancing Vertical forces: $2S \sin \beta = Q \quad (3)$

Putting value of Q from (3) in (2): $2S \sin \beta + S \sin \beta = T \cos 30 \quad (4)$

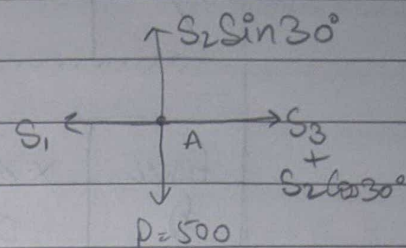
Putting value of T from (1) in (4): $3S \sin \beta = \frac{S \cos \beta}{\sin 30} \cdot \cos 30$

$$\beta = 30^\circ$$



$S_1 = Q$ as it is a same string

FBD of point A:



Force	X-dir	Y-dir
P	0	-500
S_1	$-S_1$	0
S_2	$S_2 \cos 30^\circ$	$S_2 \sin 30^\circ$
S_3	S_3	0

Balancing Vertical forces:

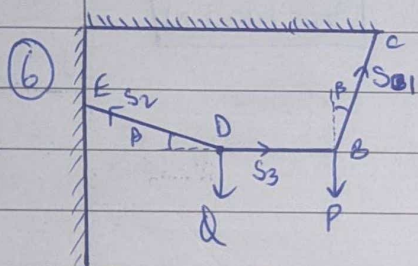
$$500 = S_2 \sin 30^\circ$$

$$S_2 = 1000 \text{ N}$$

Balancing Horizontal forces:

$$1000 = S_2 \cos 30^\circ + S_3$$

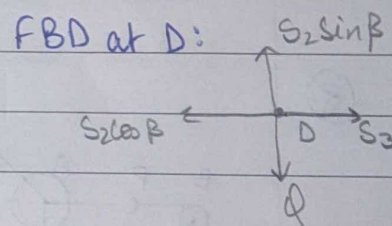
$$S_3 = 134 \text{ N}$$



$$\beta = 0.1 \text{ Radian}$$

$$\beta = 0.1 \times \frac{180^\circ}{\pi}$$

$$\beta = 5.72^\circ$$



Force	X-dir	Y-dir
Q	0	-Q
S_3	S_3	0
S_2	$-S_2 \cos \beta$	$S_2 \sin \beta$

Balancing forces in Y-dir:

$$Q = S_2 \sin 5.72^\circ$$

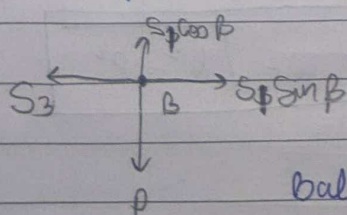
$$S_2 = 1505.01 \text{ N}$$

Balancing forces in X-dir:

$$S_3 = S_2 \cos 5.72^\circ$$

$$S_3 = 1497.5 \text{ N}$$

Balancing forces FBD of B:



Force	X-dir	Y-dir
P	0	-P
S_3	$-S_3$	0
S_1	$S_1 \cos \beta$	$S_1 \sin \beta$

Balancing Horizontal forces:

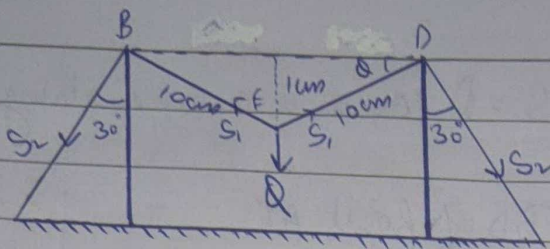
$$S_3 = S_1 \sin 5.72^\circ$$

$$S_1 = 15025.02 \text{ N}$$

Balancing Vertical forces: $S_1 \cos 5.72^\circ = P$

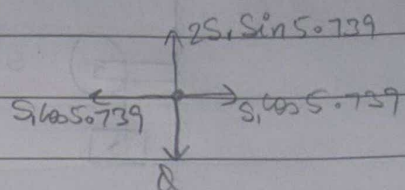
$$P = 15000 \text{ N}$$

7



$\sin \theta = 1/10$
 $\theta = 5.739$

FBD at E:



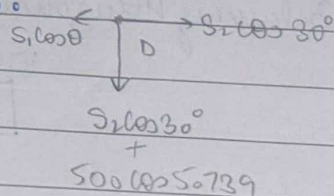
Force	x-dir	y-dir
Q	0	-Q
S ₁	$\pm S_1 \cos \theta$	$2S_1 \sin \theta$

Balancing vertical forces:

$2S_1 \sin 5.739 = 1000$

$S_1 = 500 \text{ N}$

FBD at D:

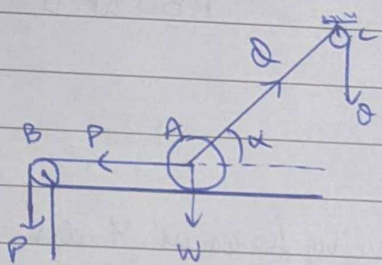


Balancing vertical forces:

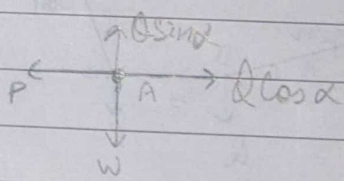
$500 \cos 5.739 = S_2 \cos 30$

$S_2 = 1000 \text{ N}$

8



FBD of A:



Forces	x-dir	y-dir
W	0	-W
P	-P	0
Q	$Q \cos \alpha$	$Q \sin \alpha$

Balancing horizontal forces:

$P = Q \cos \alpha$ - (1)

Balancing vertical forces:

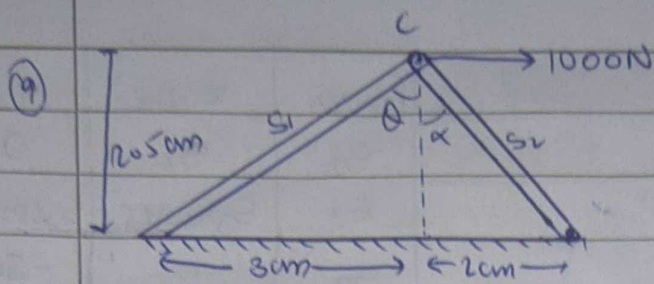
$R_h = W - Q \sin \alpha$ - (2)

$R_h = W - Q \sqrt{1 - \cos^2 \alpha}$

putting value of $\cos \alpha$ from (1)

$R_h = W - Q \sqrt{1 - \left(\frac{P}{Q}\right)^2}$

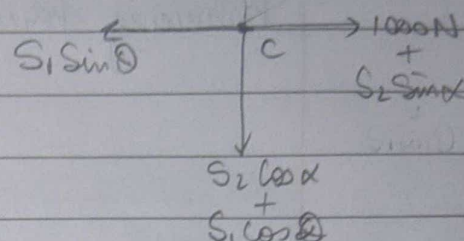
$\Rightarrow R_h = W - \sqrt{Q^2 - P^2}$



$$\tan \theta = 3/2.5 \quad \theta = 50^\circ$$

$$\tan \alpha = 2/2.5 \quad \alpha = 38.65^\circ$$

FBD at C:



Force	x dir	y dir
1000	1000	-
S_1	$-S_1 \sin \theta$	$-S_1 \cos \theta$
S_2	$-S_2 \cos \alpha$	$S_2 \sin \alpha$

Balancing vertical forces:

$$S_1 \cos \theta + S_2 \cos \alpha = 0$$

$$S_1 = -1.219 S_2 \quad \text{--- (1)}$$

Balancing Horizontal forces:

~~$$S_2 \cos \alpha = 1000 + S_1 \sin \theta$$~~

$$S_1 \sin \theta = 1000 + S_2 \sin \alpha \quad \text{--- (2)}$$

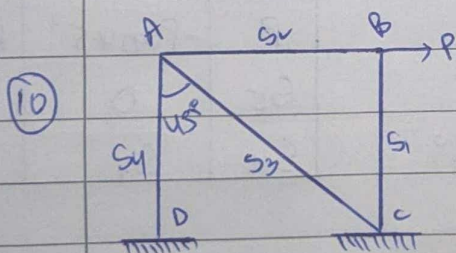
Putting value of S_1 in (2) from (1)

$$-1.219 S_2 \sin (50.19^\circ) = 1000 + S_2 \sin (38.65^\circ)$$

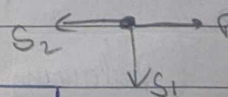
$$\boxed{S_2 = -640.65 \text{ N}}$$

Putting value of S_2 in (1): $S_1 = -1.219(-640.65)$

$$\boxed{S_1 = 782 \text{ N}}$$



FBD at point B:

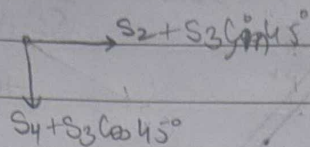


Force	x - dir	y - dir
P	P	0
S_1	0	$-S_1$
S_2	$-S_2$	0

Balancing Vertical forces: $\boxed{S_1 = 0}$

Balancing horizontal forces: $\boxed{S_2 = P}$

FBD at A:



Forces	x dir	y dir
S_2	S_2	0
S_3	$S_3 \sin 45^\circ$	$S_3 \cos 45^\circ$
S_4	0	$-S_4$

Balancing forces in Horizontal:

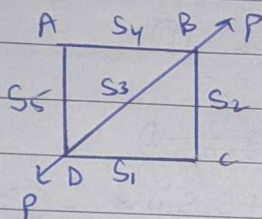
$$S_2 + S_3 \cos 45^\circ = 0$$

$$S_3 = \sqrt{2} P \quad \text{Compression}$$

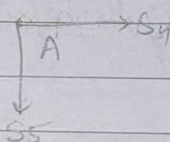
Balancing Vertical forces: $S_3 \cos 45^\circ = S_4$

$$S_4 = P$$

(11)



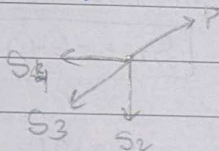
FBD at A:



Force	x dir	y dir
S_4	S_4	0
S_5	0	$-S_5$

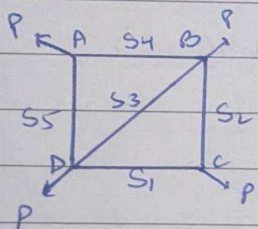
Balancing Vertical & Horizontal forces: $S_4 = 0, S_5 = 0$
 Similarly for point C, $\Rightarrow S_2 = 0, S_1 = 0$

FBD at B:

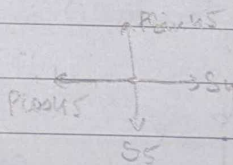


By balancing forces: $P = S_3$

(12)



FBD of A:



Forces	x dir	y dir
P	$-P \cos 45^\circ$	$P \sin 45^\circ$
S_5	0	$-S_5$
S_4	S_4	0

Balancing Vertical forces: $S_5 = P \sin 45^\circ$

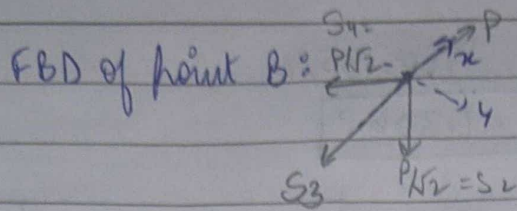
$$S_5 = \frac{P}{\sqrt{2}}$$

Balancing Horizontal forces: $S_4 = P \cos 45^\circ$

$$S_4 = \frac{P}{\sqrt{2}}$$

Similarly for point C:

$$S_1 = S_2 = \frac{P}{\sqrt{2}}$$



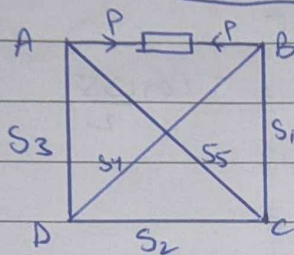
Force	x-dir	y-dir
P	P	0
S ₂	$P/\sqrt{2} \cos 45^\circ$	$P/\sqrt{2} \sin 45^\circ$
S ₄	$P/\sqrt{2} \cos 45^\circ$	$P/\sqrt{2} \sin 45^\circ$
S ₃	-S ₃	0

Balancing forces in x-dir:

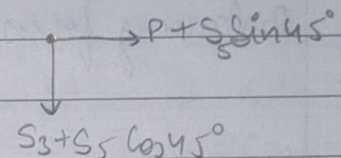
$$S_3 + P/\sqrt{2} \cos 45^\circ + P/\sqrt{2} \cos 45^\circ = P$$

$$\boxed{S_3 = P} \text{ compression}$$

(13)



FBD of point A:



Force	x-dir	y-dir
S ₃	0	-S ₃
S ₅	$S_5 \sin 45^\circ$	$S_5 \cos 45^\circ$
P	P	0

Balancing Horizontal forces:

$$S_5 \sin 45^\circ + P = 0$$

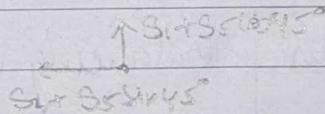
$$\boxed{S_5 = \sqrt{2} P} \text{ compression}$$

Balancing vertical forces: $S_3 + S_5 \cos 45^\circ = 0$

$$\boxed{S_3 = P} \text{ Tension}$$

Similarly for point B: $\boxed{S_1 = P \text{ \& } S_5 = \sqrt{2} P}$

FBD of point C:



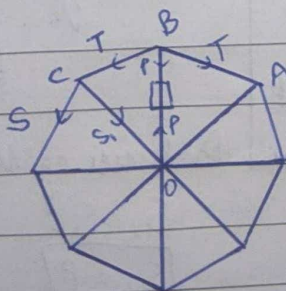
Force	x-dir	y-dir
S ₂	-S ₂	0
S ₁	0	S ₁
S ₅	$-S_5 \sin 45^\circ$	$S_5 \cos 45^\circ$

Balancing Horizontal forces:

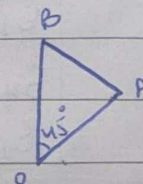
$$S_2 = S_5 \sin 45^\circ$$

$$\boxed{S_2 = P}$$

(14)

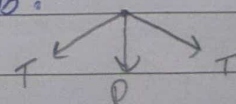


ΔOAB :



$$\angle OAB = \angle OBA = 135/2$$

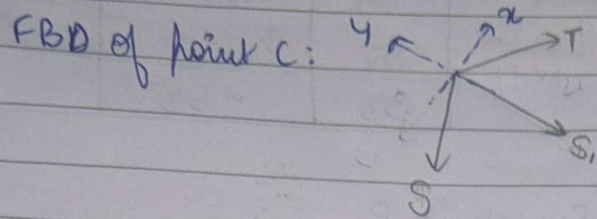
FBD of point B:



Force	x-dir	y-dir
P	-P	0
T	$-2T \cos 35^\circ$	$\pm T \sin 35^\circ$

Balancing vertical forces: $T = P \sin 135/2$

$$T = 1.306 P$$



Force	x-dir	y-dir
T	$T \sin 135/2$	$T \cos 135/2$
S	$S \sin 135/2$	$-S \cos 135/2$
S_1	0	$-S_1$

Balancing forces in x-dir: $T \sin 135/2 = S \sin 135/2$

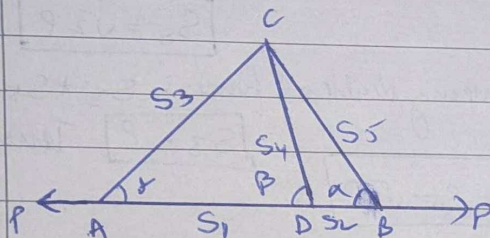
$$S = T = 1.306 P$$

Balancing forces in y-dir: $S_1 = S \cos 135/2 + T \cos 135/2$

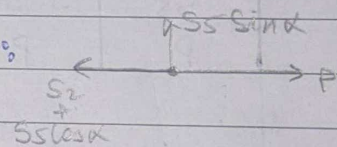
$$S_1 = P$$

Similarly for all points: all outside bars $\Rightarrow 1.306 P$
all inner bars $\Rightarrow P$

(15)



FBD of point B:

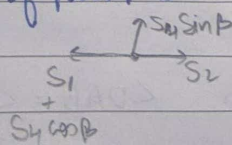


Force	x-dir	y-dir
P	P	0
S_5	$-S_5 \cos \alpha$	$+S_5 \sin \alpha$
S_2	$-S_2$	0

Balancing vertical forces: $S_5 = 0$

Balancing vertical forces: $S_2 = P$

FBD of point D:



Force	x-dir	y-dir
S_2	S_2	0
S_4	$-S_4 \cos \beta$	$S_4 \sin \beta$
S_1	$-S_1$	0

Balancing in y-dir:

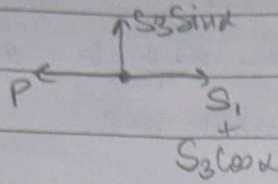
$$S_4 = 0$$

Balancing in x-dir:

$$S_1 = S_2$$

$$S_1 = P$$

FBD at point A:



Force	x-dir	y-dir
P	-P	0
S ₁	S ₁	0
S ₃	S ₃ cos α	S ₃ sin α

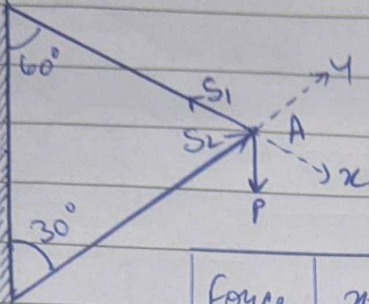
Balancing in y-dir:

$$S_3 = 0$$

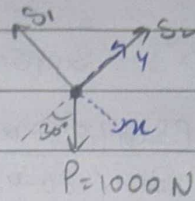
Balancing in x-dir:

$$S_1 = P$$

(16)



FBD of Point A:



Force	x-dir	y-dir
P	P sin 30°	-P cos 30°
S ₁	-S ₁	0
S ₂	0	S ₂

Balancing forces in x-dir:

$$S_1 = P \sin 30^\circ$$

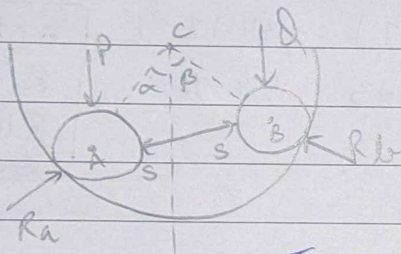
$$S_1 = 500$$

Balancing forces in y-dir:

$$S_2 = P \cos 30^\circ$$

$$S_2 = 866 \text{ N}$$

(17)



Given $P = 50 \text{ N}$, $Q = 100 \text{ N}$, $\alpha + \beta = 90^\circ$
To be found of out $\alpha = ?$, $R_a = ?$, $R_b = ?$

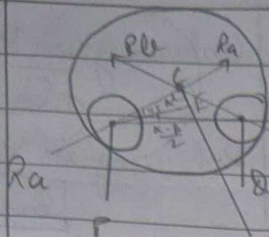
In $\triangle ABC$, $\angle ACB = 90^\circ$, $AC = BC = r$
 $\therefore \triangle ABC = \text{Right angle isosceles triangle}$

$$\angle CAB = \angle CBA = 45^\circ$$

R_a & R_b pass through $C = \text{Centre of the ring}$
 $\therefore AB$ is weight less.

$\therefore S = \text{Axial compression force acts toward A \& B}$

$\therefore Q > P \Rightarrow AB$ is not horizontal, $\alpha + \beta = 90^\circ$ given



AE ⊥ AC drawn to vertical line

$$\angle CAB = \angle CBA = 45^\circ$$

In $\triangle CEA$, $\angle CAE = B$

$$\therefore \angle EAB = 45^\circ - B = \alpha + \frac{P}{2} - B = \frac{\alpha + B - 2B}{2} = \frac{\alpha - B}{2}$$

P, Ra, S are concurrent at A.

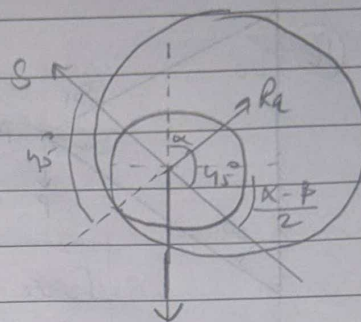
$$\frac{P}{\sin(180-45^\circ)} = \frac{S}{\sin(180-\alpha)} = \frac{Ra}{\sin(45+\alpha)}$$

$$S = \frac{P \sin(180-\alpha)}{\sin(180-45^\circ)} = \frac{P \sin \alpha}{\sin 135^\circ} = 50 \frac{\sin \alpha}{\sin 135^\circ}$$

$$S = 70.71 \sin \alpha \quad \text{--- (1)}$$

$$Ra = \frac{P \sin(45+\alpha)}{\sin 135^\circ} = 50 \frac{\sin(45+\alpha)}{\sin 135^\circ}$$

$$Ra = 70.71 \sin(\alpha + 45^\circ) \quad \text{--- (2)}$$



$$\therefore B = 90 - \alpha = 90 - 63.43 = 26.57^\circ$$

$$\frac{\alpha - B}{2} = \frac{63.43 - 26.57}{2} = 18.43^\circ$$

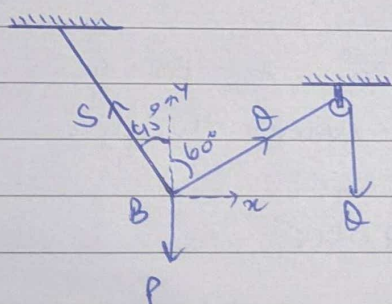
From eq (2) $Ra = 70.71 \sin(45 + 63.43) = 67.08$

From eq (1) $Rb = 141.42 \sin(45 + 26.57) = 134.17$

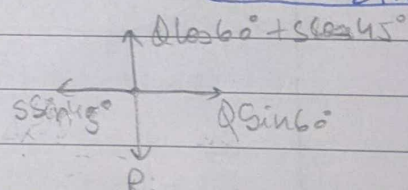
From eq (1) $S = 70.71 \sin 63.43 = 63.24$

$$\begin{aligned} \alpha - B &= 18.43^\circ \\ Ra &= 67.08 \text{ N} \\ Rb &= 134.17 \text{ N} \\ S &= 63.24 \text{ N} \end{aligned}$$

(18)



FBD of point B:



Forces	x-dir	y-dir
Q	$Q \sin 60^\circ$	$Q \cos 60^\circ$
P	0	-P
S	$S \sin 45^\circ$	$S \cos 45^\circ$

Balancing forces in x-dir:

$$S \sin 45^\circ = Q \sin 60^\circ$$

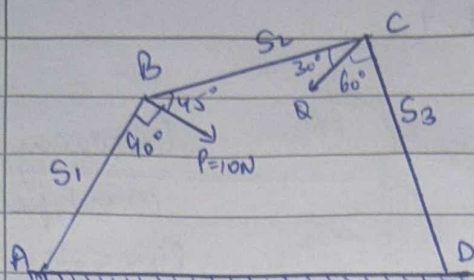
$S = 12.24 \text{ N}$

Balancing forces in y-dir:

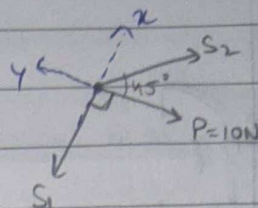
$$Q \cos 60^\circ + S \cos 45^\circ = P$$

$P = 13.65 \text{ N}$

(19)



FBD of point B:

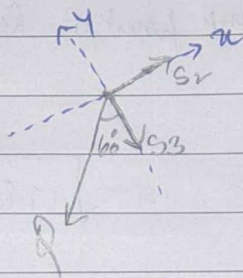


Force	x-dir	y-dir
P	0	-10
S ₁	-S ₁	0
S ₂	S ₂ sin 45°	S ₂ cos 45°

Balancing in y-dir: $P = S_2 \cos 45^\circ$

$$S_2 = \sqrt{2} P$$

FBD of point C:



Force	x-dir	y-dir
S ₂	S ₂	0
S ₃	0	-S ₃
Q	-Q sin 60°	-Q cos 60°

Balancing forces in x-dir: $Q \sin 60^\circ = S_2$ — (1)y-dir: $Q \cos 60^\circ = S_3$ — (2)

Dividing (1) by (2):

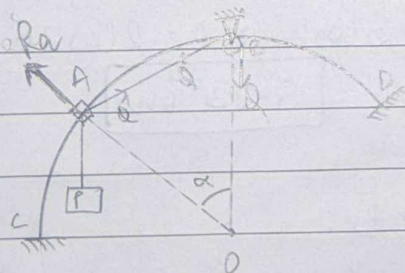
$$\tan 60^\circ = S_2 / S_3 \Rightarrow S_3 = 8.16 \text{ N}$$

Putting value of S₃ in eq (1):

$$8.16 = Q \sin 60^\circ$$

$$Q = 16.26 \text{ N}$$

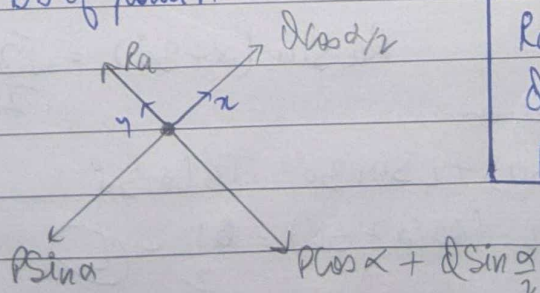
(20)

In $\triangle ABO \Rightarrow \angle OAB = \angle OBA$

$$2\alpha + \alpha = \pi$$

$$\alpha = \frac{\pi - \alpha}{2}$$

FBD of point A:



Force	x-dir	y-dir
R _a	0	R _a
Q	Q cos $\alpha/2$	-Q sin $\alpha/2$
P	-P sin α	-P cos α

Balancing in x-dir:

$$Q \cos \alpha/2 = P \sin \alpha$$

$$Q \cos \alpha/2 = 2P \sin \alpha/2 \cos \alpha/2$$

$$\frac{Q}{2P} = \sin \alpha/2$$

$$\alpha = 2 \sin^{-1} (Q/2P)$$